
CF13: THE HALF-TURN BRANCH π , CHIRALITY, AND NIELSEN–NINOMIYA PAIRING

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Abstract

This paper develops the CF13 half-turn branch of the Void Dynamics Model. The branch begins with the orthogonal re-articulation structure inherited from CF000 and CF00: a quarter-turn operator on a two-axis carrier. Its smooth completion is the ordinary rotation family. The least positive rotation that carries every state to its opposite is the half-turn constant π , and the complex expression of that same event is $e^{i\pi} = -1$. The classical transcendence of π is kept in its proper role as Lindemann’s theorem; the VDM contribution is the invariant-geometric reading that the ORS orbit leaves a footprint not captured by finite algebraic discharge. The second half of the paper follows the same half-turn structure into oriented boundaries: chirality is the effective invariant of a half-turn hosted by a five-dimensional domain wall, while Nielsen–Ninomiya pairing is the finite-periodic-closure form of the paired handed sector. The result is a compact synthesis: π , transcendence, chirality, and lattice pairing are not treated as isolated facts, but as linked appearances of one half-turn geometry. Companion materials provide optional Jupyter, SymPy, and Lean 4 support for the finite identities, numerical checks, and theorem-shaped formalization surface.

Keywords π from ORS · transcendence · chirality · Nielsen–Ninomiya · domain-wall fermions · VDM

1 Introduction

The constant π is often taught as a circle constant. Chirality is often introduced as handedness in field theory. Nielsen–Ninomiya is often introduced as a lattice no-go theorem. CF13 puts these facts on one branch.

The common structure is the half-turn. A quarter-turn first expresses orthogonal re-articulation on a two-axis carrier. Its smooth completion gives a rotation orbit. The half-turn of that orbit carries a state to its exact opposite. In complex notation this is $e^{i\pi} = -1$. In a boundary construction the same opposition becomes handedness. In finite periodic lattice closure the paired handed sector cannot be removed without leaving the hypotheses of the classical theorem.

What this paper does

The paper makes the following contributions.

1. It restates the minimal VDM inheritance used by CF13, so the reader does not need to search the whole canon to see which structures are active here.
2. It proves the local ORS-to- π chain: continuous completion, least positive half-turn, and complex half-turn closure.
3. It separates classical facts from VDM interpretation: Lindemann proves transcendence; CF13 explains why that fact fits the half-turn branch.

4. It recasts the chirality, Nielsen–Ninomiya, and domain-wall sections as formal branch statements rather than new standalone proofs of the classical theorems.
5. It gives in-paper numerical checks with explicit tolerances, negative controls, and edge cases, with companion Jupyter/SymPy/Lean materials for inspection.

Reading convention

A statement marked *Theorem* or *Proposition* is proved locally from the definitions and stated inheritance in this paper. A statement marked *Classical Theorem* is an external theorem used in its standard meaning. A *Formal Statement* or *Branch Statement* records the VDM interpretation or inherited CF08 bridge and identifies exactly what is being connected.

2 Minimal Inherited Package

This section records the local ingredients used by the paper. The goal is not to reproduce all of CF000, CF00, or CF08, but to make the CF13 proof surface visible.

Assumption 2.1 (Two-axis carrier). CF00 provides a completed real two-axis carrier $V \simeq \mathbb{R}^2$ with ordered coordinates (x, y) and the complex coordinate map

$$\chi : V \rightarrow \mathbb{C}, \quad \chi(x, y) = x + iy.$$

Assumption 2.2 (ORS quarter-turn). CF000 and CF00 provide the orthogonal re-articulation seed

$$J(x, y) = (-y, x), \quad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

so that

$$J^2 = -I, \quad J^4 = I.$$

This is the local algebraic form of the ORS quarter-turn.

Assumption 2.3 (Lawful smooth completion). The completed carrier admits the minimal smooth one-parameter completion of the ORS seed: a smooth homomorphism

$$R : \mathbb{R} \rightarrow SO(2), \quad \theta \mapsto R_\theta,$$

whose infinitesimal generator is J under the radian normalization. Equivalently, $R_\theta = \exp(\theta J)$.

Assumption 2.4 (Complex interpretation of ORS). Under $\chi(x, y) = x + iy$, the ORS seed J is multiplication by i :

$$\chi(J(x, y)) = i \chi(x, y).$$

Assumption 2.5 (CF08 domain-wall inheritance). CF08 supplies the five-dimensional auxiliary-wall construction used below: a sign-changing mass profile across an auxiliary coordinate localizes one chiral wall mode and separates the opposite partner into the far-wall or gapped sector.

Assumption 2.6 (Classical external theorems). The paper uses two classical external facts in their standard roles: Lindemann’s theorem for the transcendence of π , and the Nielsen–Ninomiya theorem under its usual locality, periodicity, translational-invariance, and chirality hypotheses.

3 Continuous ORS Completion

Theorem 3.1 (Continuous completion of ORS). *Under ?? 2.1–2.3, the ORS quarter-turn has the smooth rotational completion*

$$R_\theta = \exp(\theta J) = \cos \theta I + \sin \theta J = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

It satisfies

$$R_0 = I, \quad R_{\theta+\phi} = R_\theta R_\phi, \quad R_{\pi/2} = J, \quad R_\theta^\top R_\theta = I, \quad \det R_\theta = 1.$$

Within the normalized minimal smooth one-parameter class generated by J , this completion is unique.

Proof. Since $J^2 = -I$, the exponential series separates into even and odd powers:

$$\exp(\theta J) = \sum_{n \geq 0} \frac{\theta^{2n} J^{2n}}{(2n)!} + \sum_{n \geq 0} \frac{\theta^{2n+1} J^{2n+1}}{(2n+1)!} = \cos \theta I + \sin \theta J.$$

This gives the displayed matrix. Substituting $\theta = 0$ gives I . Since J commutes with itself,

$$R_{\theta+\phi} = \exp((\theta + \phi)J) = \exp(\theta J) \exp(\phi J) = R_\theta R_\phi.$$

The value at $\pi/2$ is

$$R_{\pi/2} = 0 \cdot I + 1 \cdot J = J.$$

The matrix is orthogonal because its columns have squared norm $\cos^2 \theta + \sin^2 \theta = 1$ and mutual inner product zero. Its determinant is $\cos^2 \theta + \sin^2 \theta = 1$. Finally, any normalized smooth one-parameter subgroup S_θ generated by J satisfies the linear initial-value problem

$$S'_\theta = S_\theta J, \quad S_0 = I.$$

The unique solution is $S_\theta = \exp(\theta J)$, so the completion is unique in the stated class. $\square$

4 The Half-Turn Constant and Euler Closure

Definition 4.1 (ORS half-turn constant). Let $\{R_\theta\}_{\theta \in \mathbb{R}}$ be the continuous ORS completion. Define

$$\pi_{\text{ORS}} := \inf\{\theta > 0 : R_\theta = -I\}.$$

When the standard radian coordinate is used, π_{ORS} is the classical π , and the notation π is used below.

Theorem 4.1 (π as the angular distance of maximal opposition). *In the standard radian coordinate, the least positive θ with $R_\theta = -I$ is $\theta = \pi$. Hence $\pi_{\text{ORS}} = \pi$, and π is the unique least positive half-turn parameter of the ORS orbit.*

Proof. By Theorem 3.1,

$$R_\theta = -I \iff \cos \theta = -1 \text{ and } \sin \theta = 0.$$

In the standard radian coordinate, the solutions are $\theta = (2m+1)\pi$ for $m \in \mathbb{Z}$. The positive solutions are $\pi, 3\pi, 5\pi, \dots$, so the least positive solution is π . Uniqueness of the least positive solution follows from the strict ordering of these positive solutions. Thus $\pi_{\text{ORS}} = \pi$. $\square$

Corollary 4.2 (Coordinate orbit, circle, and area). *For $p = (1, 0)$, the set $\{R_\theta p : 0 \leq \theta < 2\pi\}$ is the unit circle. The coordinate projections are $\cos \theta$ and $\sin \theta$, and the Euclidean area enclosed by the orbit is*

$$\iint_{x^2+y^2 \leq 1} dx dy = \pi.$$

Proof. The orbit of p is $(\cos \theta, \sin \theta)$, so it is exactly the unit circle. The enclosed region is the unit disk. Its Euclidean area is the standard polar-coordinate integral

$$\int_0^{2\pi} \int_0^1 r dr d\theta = \pi.$$

$\square$

Theorem 4.3 (Half-turn closure). *Under the complex coordinate map $\chi(x, y) = x + iy$, the continuous ORS completion acts by*

$$\chi(R_\theta(x, y)) = e^{i\theta} \chi(x, y).$$

At the half-turn,

$$e^{i\pi} = -1.$$

Proof. Using the matrix form from Theorem 3.1,

$$\chi(R_\theta(x, y)) = (x \cos \theta - y \sin \theta) + i(x \sin \theta + y \cos \theta).$$

Factoring $x + iy$ gives

$$(\cos \theta + i \sin \theta)(x + iy) = e^{i\theta} \chi(x, y).$$

At $\theta = \pi$, Theorem 4.1 gives $R_\pi = -I$, so multiplication by $e^{i\pi}$ agrees with multiplication by -1 on every complex coordinate. Applying this to $1 \in \mathbb{C}$ yields $e^{i\pi} = -1$. $\square$

5 Transcendence and Algebraic Non-Capture

Definition 5.1 (Finite algebraic closure). The algebraic numbers $\overline{\mathbb{Q}}$ are the complex numbers that solve at least one nonzero polynomial equation with rational coefficients. This is the finite algebraic closure of $\mathbb{Q}$.

Lemma 5.1 (The ORS coordinate functions are not polynomial). *The coordinate functions $\cos \theta$ and $\sin \theta$ of the continuous ORS orbit are not polynomial functions of θ .*

Proof. Both functions satisfy $f'' + f = 0$ with nonzero initial data. Let p be a nonzero polynomial of degree d . If $d \geq 1$, then p has a degree- d term, while p'' has degree at most $d - 2$; the degree- d term in $p'' + p$ cannot cancel. If $d = 0$, then $p'' + p = p$, so the only solution is $p = 0$. Therefore no nonzero polynomial solves $p'' + p = 0$. Since $\cos \theta$ and $\sin \theta$ are nonzero solutions, neither is a polynomial. $\square$

Classical Theorem 5.2 (Lindemann). *The classical circle constant π is transcendental:*

$$\pi \notin \overline{\mathbb{Q}}.$$

Proof. Lindemann's theorem states that if α is a nonzero algebraic number, then e^α is transcendental. Suppose for contradiction that π were algebraic. Then $i\pi$ would be a nonzero algebraic number. Lindemann's theorem would imply that $e^{i\pi}$ is transcendental. But Theorem 4.3 gives $e^{i\pi} = -1$, which is algebraic. This contradiction proves that π is transcendental. $\square$

Proposition 5.3 (VDM interpretation of transcendence). *Within CF13, the transcendence of π says that the numerical footprint of the ORS half-turn is not exhausted by finite algebraic closure. The two-axis carrier can host the orbit, but the least opposite-pole distance belongs to the completed geometric orbit rather than to finite polynomial discharge.*

Proof. By Theorems 3.1 and 4.1, π is fixed by the completed ORS orbit as the first opposite-pole parameter. By Theorem 5.2, this value is not algebraic. Therefore the half-turn is visible on the algebraic carrier but not reducible to a finite algebraic number. This is exactly the CF13 reading: the orbit's geometric completion leaves a numerical trace that finite algebraic closure does not capture. $\square$

6 Formal Branch Statements

Definition 6.1 (Half-turn witness). For a nonzero carrier state $z \in \mathbb{C}$, the half-turn witness is the operation

$$z \mapsto e^{i\pi} z = -z.$$

It sends each state to its exact opposite on the same carrier orbit.

Proposition 6.1 (Maximal geometric opposition). *On the unit ORS orbit, the half-turn is the unique least positive rotation that reaches the antipodal state. It is therefore the carrier-level witness of maximal geometric opposition.*

Proof. This is the geometric restatement of Theorem 4.1. On the unit orbit, antipodal points are exactly related by multiplication by -1 . The least positive rotation realizing that map is π . $\square$

Formal Statement 6.2 (Chirality from the CF08 boundary). In the CF08 auxiliary-wall construction, half-turn opposition is expressed as an oriented boundary. The effective invariant selected at that boundary is chirality.

Justification. By Assumption 2.5, the inherited construction uses a sign-changing mass profile across an auxiliary coordinate. The wall separates opposite sign sectors and localizes one chiral mode at the interface. A sign choice across an oriented interface selects a handed sector. Thus the boundary expression of half-turn opposition is read as chirality in the effective theory. $\square$

Branch Statement 6.3 (Nielsen–Ninomiya as finite closure pairing). Under the standard Nielsen–Ninomiya hypotheses, a single isolated chiral sector cannot be retained on a finite periodic lattice without the paired partner. In the CF13 reading, this is the finite-periodic-closure expression of half-turn pairing.

Justification. The classical theorem supplies the pairing conclusion under its stated hypotheses. CF13 identifies a chiral sector as the oriented-boundary expression of half-turn opposition. Finite periodic closure removes the asymptotic direction that would isolate one handed sector, so the paired sector returns inside the closed lattice description. This is an interpretation of the classical theorem, not a new proof replacing it. $\square$

Branch Statement 6.4 (Domain-wall separation of the partner). The CF08 domain-wall construction separates the paired partner along the auxiliary direction rather than deleting it.

Justification. By Assumption 2.5, one chiral zero mode localizes at the wall, while the opposite partner is displaced into the far-wall or gapped sector. The enlarged host keeps the pairing logic intact, but the low-energy physical slice sees one localized handed mode. This is the branch-level reason domain walls help: they change the host geometry of the partner rather than deny its existence. $\square$

7 Numerical Validation

The numerical checks in this section are small enough to verify independently. They are not substitutes for the proofs above; they make the finite identities, tolerances, and controls visible. The declared double-precision tolerance is

$$\tau_{\text{num}} = 10^{-12}.$$

The rotation grid uses 73×73 pairs (a, b) with $a, b \in [-10\pi, 10\pi]$. The distorted-rotation control uses

$$\tilde{R}_\theta = \begin{pmatrix} \cos \theta & -1.05 \sin \theta \\ \sin \theta & \cos \theta \end{pmatrix},$$

which resembles a rotation but is not a one-parameter subgroup of $SO(2)$.

Table 1: Hard-checkable finite validation values. Positive checks sit below $\tau_{\text{num}} = 10^{-12}$; controls are deliberately far above the tolerance or deliberately violate the relevant hypothesis.

Quantity	Method	Value	Reading
Composition law	$\max \ R_{a+b} - R_a R_b\ _F$	5.18×10^{-15}	$< \tau_{\text{num}}$
Orthogonality	$\max \ R_a^\top R_a - I\ _F$	3.20×10^{-16}	$< \tau_{\text{num}}$
Half-turn	$\ R_\pi + I\ _F$	1.73×10^{-16}	$< \tau_{\text{num}}$
Euler closure	$ e^{i\pi} + 1 $	1.22×10^{-16}	$< \tau_{\text{num}}$
Large-angle edge case	$\ R_{20\pi} - I\ _F$	3.46×10^{-15}	$< \tau_{\text{num}}$
Distorted rotation control	$\max \ \tilde{R}_{a+b} - \tilde{R}_a \tilde{R}_b\ _F$	7.07×10^{-2}	control
Off-target control	$\ R_{\pi/2} + I\ _F$	2.00	control
Periodic pairing toy	slopes of $\sin k$ at $0, \pi$	$+1, -1$	paired
Nonperiodic single-cone control	$D(k) = k: D(\pi) - D(-\pi) $	2π	not periodic
Domain-wall envelope	$\ \text{sech}(z)/\sqrt{2}\ _{L^2}^2$	1.000000	normalized

The companion notebook repeats these calculations, includes the symbolic simplifications, and gives two small visual checks: the periodic pairing toy model and the localized domain-wall envelope. The SymPy audit emits the same metrics as JSON.

8 Scope and Companion Materials

The current paper proves the local ORS-to- π chain from the stated inherited package. It does not reprove all of CF000 or CF00. It uses Lindemann’s theorem as a classical theorem, not as a VDM-derived theorem. It uses Nielsen–Ninomiya as a classical theorem, not as a replacement theorem. It uses the CF08 domain-wall construction as inherited structure and states exactly how CF13 reads that structure.

Companion materials provide optional support:

- a Jupyter notebook with symbolic checks, numerical validation, controls, and notebook figures;
- a SymPy audit script producing machine-readable metrics;
- a Lean 4 theorem-shaped surface for finite state-completeness and quarter-memory statements;

- a provenance manifest and publication standards documents.

These materials are not needed to follow the main argument, but they make the finite checks easy to rerun and extend.

9 Conclusion

CF13 identifies a single half-turn branch connecting π , complex phase closure, transcendence, chirality, and Nielsen–Ninomiya pairing. The ORS quarter-turn continuously completes to the rotation family. The least positive opposite-pole parameter is π . The complex expression of that half-turn is $e^{i\pi} = -1$. Lindemann’s theorem then locates π outside finite algebraic closure, while VDM explains why this belongs to the geometry of the completed orbit. When the same half-turn opposition is hosted by an oriented auxiliary boundary, it becomes chirality; when finite periodic closure is imposed, the paired handed sector returns as Nielsen–Ninomiya pairing. Domain walls succeed by separating the partner in a larger lawful host.

A Self-Contained Appendix: Minimal CF13 Structures

This appendix restates the proof-critical structures in compact form.

- S1. Carrier.** CF13 works on a two-axis real carrier $V \simeq \mathbb{R}^2$ with complex readout $\chi(x, y) = x + iy$.
- S2. Quarter-turn.** The ORS seed is $J(x, y) = (-y, x)$, so $J^2 = -I$ and $J^4 = I$.
- S3. Smooth completion.** The lawful continuous completion is the normalized smooth one-parameter subgroup generated by J : $R_\theta = \exp(\theta J)$.
- S4. Half-turn.** The half-turn constant is the least positive θ with $R_\theta = -I$; in standard radian coordinates this is π .
- S5. Complex action.** The complex readout turns R_θ into multiplication by $e^{i\theta}$.
- S6. Classical transcendence.** Lindemann proves $\pi \notin \overline{\mathbb{Q}}$.
- S7. Boundary branch.** CF08 supplies the auxiliary-wall mechanism used to read oriented half-turn opposition as chirality.
- S8. Finite closure branch.** Nielsen–Ninomiya supplies the classical finite periodic lattice pairing result.

B Numerical Details

The companion audit computes the following symbolic identities:

$$R_{\theta+\phi} - R_\theta R_\phi = 0, \quad R_\pi + I = 0, \quad e^{i\pi} + 1 = 0.$$

For the grid calculations, the Frobenius norm is used. The off-target control $\|R_{\pi/2} + I\|_F = 2$ demonstrates that the quarter-turn is not mistaken for a half-turn. The distorted control demonstrates that a visually rotation-like matrix fails the group law when it is not the ORS completion.

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